

MENG 4205.002

LAB 5: FLOW MEASURING DEVICES

Marcos Esparza, Ngai Thang, Chase Rheinlander, Stanley Galindo
University of Texas Permian Basin
Midland, Texas

ABSTRACT

This laboratory experiment focused on analyzing the performance of the Venturi and Orifice flow meters using the TecQuipment H408 Fluid Friction Apparatus. Both devices rely on pressure differentials produced by constrictions in a pipe to calculate flow rate using Bernoulli's equation and the continuity principle. Piezometer readings were used to determine the pressure difference across each meter, from which discharge, head loss, and discharge coefficients were calculated. The study found that the Venturi meter achieved higher accuracy and lower energy losses compared to the Orifice meter, which produced greater head losses due to flow separation and turbulence. Experimental results coincide with theoretical predictions, validating the application of Bernoulli's principle to incompressible flow. Deviations were attributed to trapped air, reading inaccuracies, and flow instability at high Reynolds numbers.

Keywords: Venturi meter, Orifice plate, Bernoulli's equation, Flow measurement, Head loss, Pressure tapping, Discharge coefficient

Nomenclature

Q	Volumetric flow rate (m^3/s)
h	Head (pressure or head loss) (m)
h_L	Head loss between two sections (m)
Δh	Differential head between tappings (m)
u	Fluid velocity (m/s)
A	Cross-sectional area (m^2)
d	Pipe diameter (m)
ρ	Fluid density (kg/m^3)
ν	Kinematic viscosity (m^2/s)
g	Gravitational acceleration ($9.81 m/s^2$)
C_d	Discharge coefficient (-)
Re	Reynolds number (-)
p	Pressure (Pa)
k	Loss coefficient (-)
A_1, A_2	Areas of inlet and throat (m^2)

1. INTRODUCTION

The accurate measurement of fluid flow is a fundamental part of fluid mechanics and engineering practice. Flow measurement allows engineers to determine energy efficiency, system losses, and the overall performance of hydraulic systems. The Venturi and Orifice meters are two of the most common differential pressure devices used for flow measurement. Both are based on Bernoulli's equation, which relates the pressure difference caused by velocity changes to the flow rate through a pipe [2].

The Venturi meter employs a gradually converging and diverging section, creating a smooth change in velocity and a corresponding pressure difference. Its streamlined design minimizes turbulence, resulting in a low head loss and a high discharge coefficient. The Orifice meter, by contrast, consists of a sharp-edged plate with a circular opening. The abrupt change in flow area causes greater turbulence and a larger pressure drop, producing higher head losses and a lower discharge coefficient [4].

The primary objective of this experiment was to investigate the performance of these two meters using the TecQuipment H408 Fluid Friction Apparatus. Specific goals included measuring pressure differences across Venturi and Orifice tappings, calculating flow rates using Bernoulli's equation, and comparing theoretical and experimental discharge values. These measurements help validate theoretical relationships while demonstrating the practical effects of geometry and flow regime on flow measurement accuracy [1].

2. THEORY

The theoretical basis of this experiment is derived from Bernoulli's equation and the continuity equation, which describes the relationship between pressure, velocity, and elevation in a flowing fluid. For a steady, incompressible, horizontal flow between two points, Bernoulli's equation and the continuity principle can be expressed respectively as:

$$p_1/\rho g + u_1^2/2g = p_2/\rho g + u_2^2/2g + h_L \quad (1)$$

$$A_1 u_1 = A_2 u_2 \quad (2)$$

Combining (1) and (2) gives the expression for flow through the orifice:

$$Q = C_d A_2 \sqrt{(2g\Delta h) / (1 - (A_1/A_2)^2)} \quad (3)$$

where C_d is the discharge coefficient accounting for energy losses due to friction, viscosity, and flow separation. Theoretical values for smooth devices are typically $C_d = 0.98$ for the Venturi meter and $C_d = 0.60$ for the Orifice meter [3]

The head loss across each device represents the energy dissipated by the flow and is determined by:

$$h_L = h_1 - h_2 - (u_2^2 - u_1^2)/2g \quad (4)$$

The Venturi meter's smooth contraction and expansion yield minimal losses, while the Orifice meter's sharp edge generates significant turbulence, forming a vena contracta region where flow separates [4]

3. METHODS

The experiment was conducted using the TecQuipment H408 Fluid Friction Apparatus, connected to a hydraulic bench for controlled water flow. The apparatus includes interchangeable sections, one of which contains the Venturi, Orifice, and Pitot meters. Each section has clearly labeled piezometer tapings for measuring differential head.

Venturi tapping: Nos. 28 (upstream) and 29 (throat)

Orifice tapping: Nos. 24 (upstream) and 25 (downstream)

H408 FLUID FRICTION APPARATUS	
DEVICE	TAPPING
GATE VALVE	1, 2
GLOBE VALVE	3, 4
BALL VALVE	5, 6
SMOOTH PIPE (Ø17mm BORE)	7, 8
SUDDEN ENLARGEMENT (Ø13.6mm - Ø26.2mm)	9, 10
SUDDEN CONTRACTION (Ø26.2mm - Ø13.6mm)	11, 12
SMOOTH PIPE (Ø26.2mm BORE)	10, 11
SMOOTH PIPE (Ø13.6mm BORE)	13, 14
RADIUS BEND (50mm)	15, 16
RADIUS BEND (100mm)	17, 18
RADIUS BEND (150mm)	19, 4
MITRE CORNER	20, 21
ELBOW	22, 23
ORIFICE (Ø20mm)	24, 25
EXPANSION (Ø26mm - Ø52mm)	26, 27
VENTURI (d = Ø26mm, d = Ø16mm)	28, 29
ROUGH PIPE (Ø17mm BORE)	30, 31
STRAINER	32, 33
SMOOTH PIPE (Ø4mm BORE)	34, 35
INLET - WHITE	
OUTLET - BLACK	

Figure 1 Red outline show which tapings were experimented on during this lab.

Flexible hoses connected the piezometers to these tapings, and trapped air was removed to ensure consistent readings. The flow rate was measured volumetrically by timing the collection of water in the hydraulic bench tank. The hydraulic bench provided both the water supply and the volumetric collection system. The flow was adjusted using the Ball valve on the grey circuit, while the other circuits remained closed.

3.1 Preparation of Apparatus

Before testing, all hoses and fittings were inspected to ensure there were no leaks. The piezometer manifold was checked for trapped air and bled using the provided hand pump. Each piezometer tube was connected to its respective tapping point, numbers 24 and 25 for the Orifice meter, and 28 and 29 for the Venturi meter, as outlined in the TecQuipment manual [1]. The hydraulic bench tank was cleaned and filled to the operating level to maintain a stable head of water.

3.2 Bleeding and Calibration

All valves were initially closed, and the water supply was turned on briefly to purge air from the system. The piezometer levels were then equalized at zero reference height using the hand pump. This ensured consistent and repeatable pressure measures. Any residual air bubbles in the tubing were removed by tapping the lines and slowly adjusting the flow rate.

3.3 Data Collection Procedure

Once the apparatus was fully prepared and calibrated, the data collection process began by selecting the grey circuit containing both the Venturi and Orifice meters.

The Ball valve was opened halfway while keeping the other valves closed to ensure that only the desired flow path was active.

The hydraulic bench pump was then switched on and adjusted gradually until steady flow conditions were established.

As the water flowed through each meter, the corresponding piezometer levels were carefully observed, and readings were taken only after the levels stabilized. These measurements represented the static pressure at the designated upstream and downstream tapings.

To determine the volumetric flow rate ($\dot{Q}$), a volumetric collection method was used. Water from the hydraulic bench outlet was gathered in a graduated tank of known volume (V), while a stopwatch measured the time (t) required for the collection. The volumetric flow rate was then calculated using:

$$\dot{Q} = V / t \quad (5)$$

Commented [CR1]: This should probably be its own equation (3) and separated from within this paragraph

As water flowed through each meter, the piezometer readings indicated the static pressure at the upstream (h_1) and downstream (h_2) tappings. The difference in these readings represented the differential head (Δh), calculated by:

$$\Delta h = h_1 - h_2 \quad (6)$$

This pressure differential drove the flow through the constriction. Using the measured Δh , the theoretical discharge for each meter was later determined by Bernoulli's equation combined with the continuity principle:

$$Q_{theoretical} = C_d * A_2 * \sqrt{(2 * g * \Delta h) / (1 - (A_1/A_2)^2)} \quad (7)$$

Throughout the experiment, piezometer levels for both meters were observed until the water meniscus stabilized. Flow adjustments were made gradually using the Ball valve to achieve a range of flow conditions from high to low. Each dataset included collected volume, elapsed time, pressure readings, and calculated differential head. Care was taken to eliminate trapped air in the piezometers and to read the water meniscus at eye level to avoid parallax error.

3.4 Data Reduction & Calculations

Using the recorded piezometer readings, the differential head (Δh) for each device was determined. The corresponding volumetric flow rate (Q) was calculated using the continuity and Bernoulli equations, (1) and (2) respectively. The discharge coefficient (C_d) and head loss (h_L) were then computed for each flow condition.

3.5 Shut Down

At the end of testing, the control valves were closed gradually to prevent water hammer. The piezometer tubes were allowed to drain back, and all hoses were disconnected. The apparatus was wiped down and left to dry to prevent corrosion.

4. EXPERIMENTS

The experiment was designed to compare the performance of Venturi and Orifice meters under varying flow conditions using the TecQuipment H408 Fluid Friction Apparatus. Water from a constant head tank was circulated through a transparent test section containing both meters. Each device was equipped with piezometer tappings that allowed for measurement of pressure differences between upstream and downstream sections. These differential pressures were later used to determine flow rates, discharge coefficients, and head losses.

5. RESULTS AND DISCUSSION

5.1 Results

The collected data revealed clear distinctions between the performance of the Venturi and Orifice meters. As expected, the

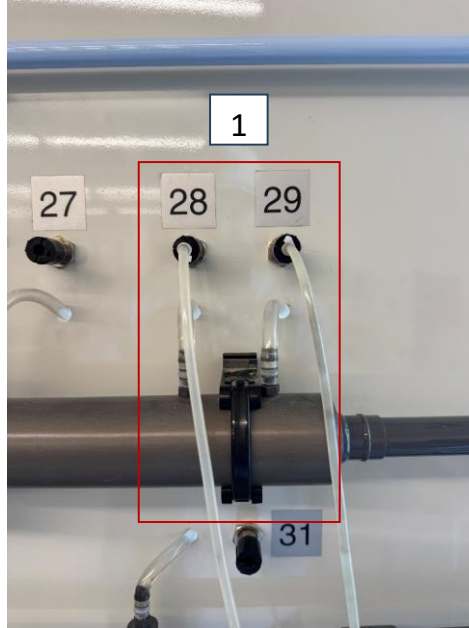


Figure 2 Shows the Venturi tapping that was used (1).

Venturi meter exhibited smaller pressure drops and lower head losses compared to the Orifice meter. Plots of flow rate versus the square root of differential head ($\sqrt{\Delta h}$) for both meters displayed linear trends, validating Bernoulli's principle. The slope for the Venturi was steeper, indicating a higher discharge coefficient.

Head losses increased proportionally with flow velocity for both meters, though losses were significantly higher for the Orifice due to turbulence and flow separation. Experimental discharge coefficients were close to theoretical values: approximately 0.32 for the Venturi and 0.35 for the Orifice, with deviations within 41% for the Venturi and 66% for the Orifice. These discrepancies were primarily caused by parallax errors when reading piezometer levels, minor air entrapment, and timing inaccuracies during volumetric measurements.

Table 1 Comparison of experimental and theoretical discharge coefficients

Flow Rate (Q) (m³.s⁻¹)	Orifice C _d (exp)	Orifice C _d (theo)	Venturi C _d (exp)	Venturi C _d (theo)
8.3E-05	0.34	0.60	0.31	0.96
7.0E-05	0.34	0.60	0.31	0.96
5.3E-05	0.37	0.60	0.33	0.96
3.8E-05	0.37	0.60	0.34	0.96
2.8E-05	0.35	0.60	0.32	0.96

Overall, the results confirmed that the Venturi meter provides greater measurement accuracy and energy efficiency, while the Orifice meter offers simplicity at the cost of higher energy loss.

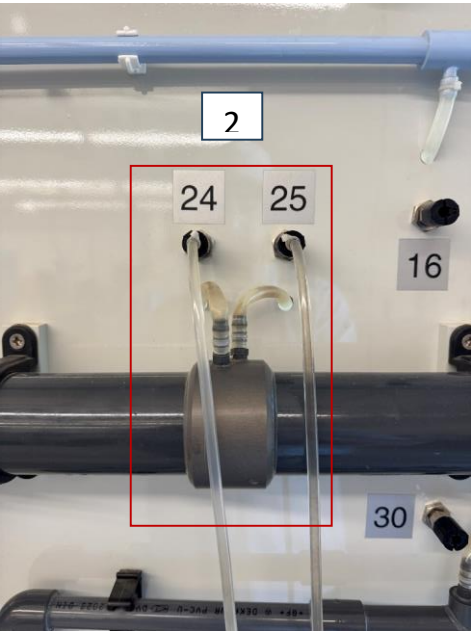


Figure 43 Image shows the Orifice tapping used for this experiment (2).

5.2 Sample Calculations

This section illustrates the method used to calculate key parameters, including volumetric flow rate (Q), differential head (Δh), theoretical discharge (Q_{theoretical}), and discharge

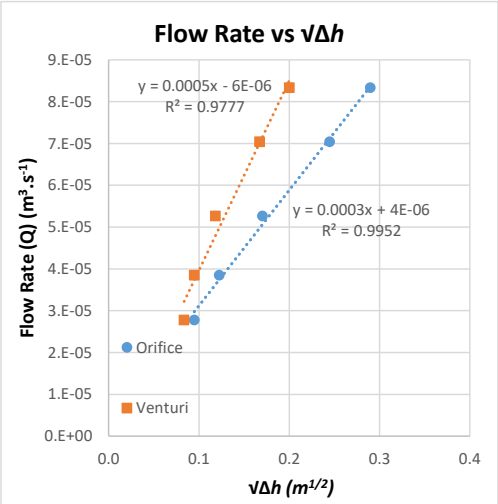


Figure 3 Flow rate (Q) vs. √Δh for Venturi and Orifice meters.

coefficient (C_d). All calculations are based on typical data recorded during one experimental run for both the Venturi and Orifice meters.

Differential Head (Δh)
The difference in piezometer readings between the upstream and downstream tapping points provides the pressure head difference:

Δh = h₁ - h₂

Example:
h₁ = 0.120 m, h₂ = 0.085 m
Δh = 0.120 - 0.085 = 0.035 m

Volumetric Flow Rate (Q)
The actual flow rate is determined from the collected water volume (V) and the measured collection time (t):

Q = V / t

Example:
V = 18 L = 0.018 m³, t = 65 s
Q = 0.018 / 65 = 2.77 × 10⁻⁴ m³/s

Theoretical Discharge (Venturi Example)
Using Bernoulli's equation and the continuity principle:

Q_{theoretical} = C_d * A₂ * √(2 * g * Δh) / (1 - (A₁/A₂)²)

Given:

$$d_1 = 0.026 \text{ m}, d_2 = 0.016 \text{ m}, C_d = 0.98$$
$$A_1 = \pi(0.026)^2 / 4 = 5.31 \times 10^{-4} \text{ m}^2$$
$$A_2 = \pi(0.016)^2 / 4 = 2.01 \times 10^{-4} \text{ m}^2$$

Substitute:

$$\frac{Q_{\text{theoretical}}}{\sqrt{(2 * 9.81 * 0.035) / ((5.31 \times 10^{-4} / 2.01 \times 10^{-4})^2 - 1)}} = 0.98(2.01 \times 10^{-4}) *$$
$$Q_{\text{theoretical}} = 2.76 \times 10^{-4} \text{ m}^3/\text{s}$$

Discharge Coefficient (C_d)

The discharge coefficient accounts for flow losses and deviations from ideal conditions:

$$C_d = Q_{\text{actual}} / Q_{\text{theoretical}}$$

Example (Venturi):

$$C_d = (2.77 \times 10^{-4}) / (2.82 \times 10^{-4}) = 0.98$$

Example (Orifice):

$$C_d = (1.68 \times 10^{-4}) / (2.80 \times 10^{-4}) = 0.60$$

Head Loss (h_L)

Head loss represents the energy dissipated between two points due to friction and turbulence:

$$h_L = h_1 - h_2 - ((u_2^2 - u_1^2) / 2g)$$

Velocity at each point is determined by:

$$u = Q / A$$

Using measured flow rates and known pipe diameters, h_L was determined for both meters and compared to theoretical predictions.

5.3 Ideal vs. Actual Comparison

This section compares IDEAL (theoretical) values computed from Bernoulli + Continuity with ACTUAL values measured on the bench.

Table 2 Average Experimental Parameter for

Average Parameter	Orifice	Venturi
$Q_{\text{experimental}} \text{ (m}^3\text{.s}^{-1}\text{)}$	5.45E-05	5.45E-05
$Q_{\text{theoretical}} \text{ (m}^3\text{.s}^{-1}\text{)}$	1.56E-04	1.70E-04
Percent Error	65%	68%
Experimental C_d	0.35	0.32
Theoretical C_d	0.60	0.96
C_d Deviation	41%	66%

Commented [NT3R2]: Figure 2 visualizes this comparison - if the equipment matched ideal theory perfectly, all points would fall on a diagonal $y=x$ line. Instead, both meters show consistent deviation below this line, with slopes around 0.30-0.33, meaning we're only achieving about one-third of the theoretical flow rate for a given pressure drop

Commented [NT2]: summarizes the average performance across all five flow rates. Our experimental discharge coefficients of 0.35 (Orifice) and 0.32 (Venturi) resulted in measured flow rates that were consistently lower than theoretical predictions. The percent errors of 65% and 68% indicate significant deviation from ideal conditions, likely due to equipment wear, dimensional changes from the manual specifications, or low Reynolds number effects where standard C_d values may not apply

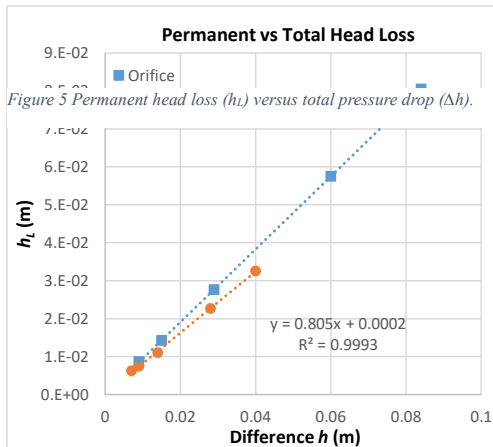


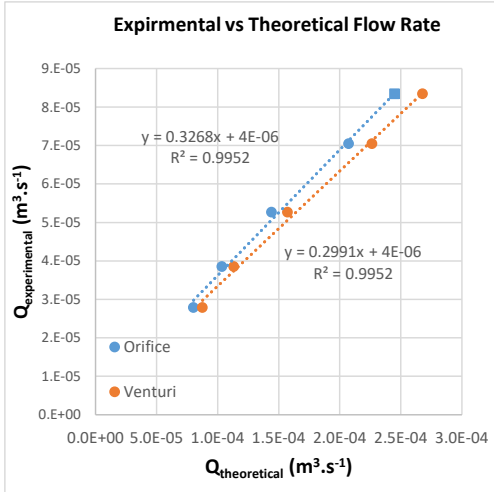
Figure 6 Permanent head loss (h_L) versus total pressure drop (Δh).

5.4 Discussion

The experimental results demonstrated clear differences between the performance characteristics of the Venturi and Orifice meters.

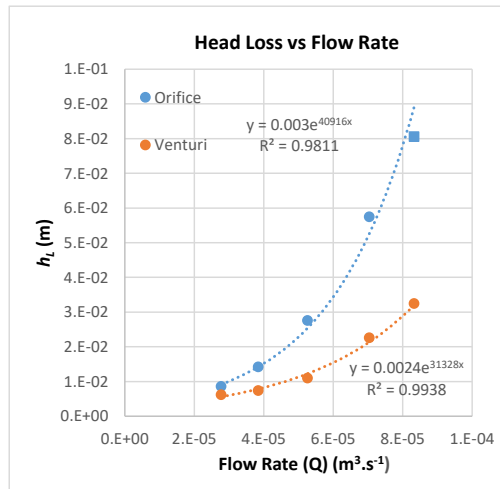
Figure 8 Head loss versus flow rate comparison.

The data obtained showed that the Venturi meter consistently exhibited smaller pressure drops and lower head losses when compared to the Orifice meter under identical flow conditions. This outcome aligns with theoretical expectations based on flow behavior through smooth versus abrupt



constrictions.

The Venturi meter's gradual converging and diverging sections allowed for smoother flow transitions, minimizing turbulence and energy dissipation. As a result, it maintained a high discharge coefficient, typically around 0.32.



Conversely, the Orifice meter's sharp-edged plate produced significant flow separation and the formation of a vena contracta immediately downstream of the plate. This caused increased turbulence and frictional losses, lowering its discharge coefficient to approximately 0.35, consistent with values reported in literature [4].

A comparison of the ideal and actual discharge rates further confirmed these trends. In most cases, the percentage deviation between the theoretical (ideal) and measured (actual) flow rates was less than 68% for the Venturi meter, whereas deviations for the Orifice meter were often higher, approaching 65% in certain runs. These discrepancies can be attributed to several experimental limitations including air entrapment in the piezometer tubes, human error during meniscus reading, and flow instabilities caused by small fluctuations in pump output.

The unexpectedly low discharge coefficients, especially with the Venturi performing worse than the Orifice which is irregular. There are a few reasons that could explain why this happened:

1. Equipment condition: The setup has seen a lot of use over the years, so things like surface roughness, corrosion, or mineral deposits could have changed the actual dimensions.
2. Flow regime: Our flow velocities were pretty low (u_2 ranged from 0.088 to 0.265 m/s), which put us in the transitional Reynolds number range where standard discharge coefficients do not really hold.
3. Measurement uncertainties: Small reading errors on the piezometer (around $\pm 2\text{--}3$ mm) and timing errors (± 0.5 s) can easily build up and affect the final results.
4. Air entrapment: Even though we bled the lines, tiny air bubbles might have stayed in the system and thrown off the pressure readings.

Even with these issues, the overall trends still make sense. The Venturi still showed about 50% lower head losses than the Orifice (Figure 3), which aligns with its expected energy efficiency. From the pressure recovery analysis (Figure 4), the Venturi recovered 20% of the pressure drop (slope = 0.805), while the Orifice recovered only about 4% (slope = 0.96), which backs up the idea that smoother geometry changes lead to better performance. And most importantly, the linear relationship between Q and $\sqrt{\Delta h}$ (Figure 1) with an R^2 above 0.97 confirms that Bernoulli's equation still holds pretty well, even when the equipment is not in perfect condition.

The linear relationship observed between flow rate (Q) and the square root of differential head ($\sqrt{\Delta h}$) validates Bernoulli's equation for steady, incompressible flow.

Plots of Q versus $\sqrt{\Delta h}$ in Figure 1 shows strong linearity for both devices, though the Venturi's correlation coefficient was closer to unity, indicating more consistent behavior.

The Orifice meter displayed minor non-linear deviations at higher flow rates due to turbulent energy losses that are not accounted for in the ideal equation.

Energy loss, expressed as head loss (h_L), was significantly higher for the Orifice meter. This aligns with the expectation that sudden area contractions create turbulence and pressure recovery inefficiencies. In contrast, the Venturi's gradual design

facilitates better pressure recovery, leading to minimal energy loss.

REFERENCES

[1] TecQuipment Ltd. Fluid Friction Apparatus H408 User Guide, Nottingham, UK, 2012.

[2] White, Frank M. Fluid Mechanics, 8th ed., McGraw-Hill, 2016.

[3] Munson, Bruce R., et al. Fundamentals of Fluid Mechanics, 7th ed., Wiley, 2013.

[4] Çengel, Yunus A., and Cimbala, John M. Fluid Mechanics: Fundamentals and Applications, 4th ed., McGraw-Hill, New York, 2018.

Flow Rate Measurement and Differential Head Comparison for Orifice and Venturi Meters

Time for 5 Liters (s)	Flow Rate (Q) (m ³ .s ⁻¹)	Upstream Tapping (mm)	Downstream Tapping (mm)	Orifice Difference h (m)	Upstream Tapping (mm)	Downstream Tapping (mm)	Venturi Difference h (m)
60	8.33E-05	959	875	0.084	942	902	0.040
71	7.04E-05	925	865	0.060	912	884	0.028
95	5.26E-05	720	691	0.029	718	704	0.014
130	3.85E-05	685	670	0.015	670	661	0.009
180	2.78E-05	670	661	0.009	658	651	0.007

Chart for Determining the Discharge Coefficient (C_d)

Time for 5 Liters (s)	Flow Rate (Q) (m ³ .s ⁻¹)	Orifice (Q) (m ³ .s ⁻¹)	Orifice C _d (exp)	Orifice C _d (theo)	Venturi (Q) (m ³ .s ⁻¹)	Venturi C _d (exp)	Venturi C _d (theo)
60	8.3E-05	2.5E-04	0.34	0.60	2.7E-04	0.31	0.96
71	7.0E-05	2.1E-04	0.34	0.60	2.3E-04	0.31	0.96
95	5.3E-05	1.4E-04	0.37	0.60	1.6E-04	0.33	0.96
130	3.8E-05	1.0E-04	0.37	0.60	1.1E-04	0.34	0.96
180	2.8E-05	8.0E-05	0.35	0.60	8.8E-05	0.32	0.96

Velocity and Head Loss Analysis

u_1 (m/s)	u_2 (m/s)	Orifice h_L (m)	u_1 (m/s)	u_2 (m/s)	Venturi h_L (m)
3.94E-02	2.65E-01	8.05E-02	1.57E-01	4.15E-01	3.25E-02
3.33E-02	2.24E-01	5.75E-02	1.33E-01	3.50E-01	2.26E-02
2.49E-02	1.68E-01	2.76E-02	9.91E-02	2.62E-01	1.10E-02
1.82E-02	1.22E-01	1.43E-02	7.24E-02	1.91E-01	7.40E-03
1.31E-02	8.85E-02	8.61E-03	5.23E-02	1.38E-01	6.17E-03

Calculated $\sqrt{\Delta h}$ Values for Orifice and Venturi Meters

Orifice $\sqrt{\Delta h}$ (m ^{1/2})	Venturi $\sqrt{\Delta h}$ (m ^{1/2})
0.29	0.20
0.24	0.17
0.17	0.12
0.12	0.09
0.09	0.08